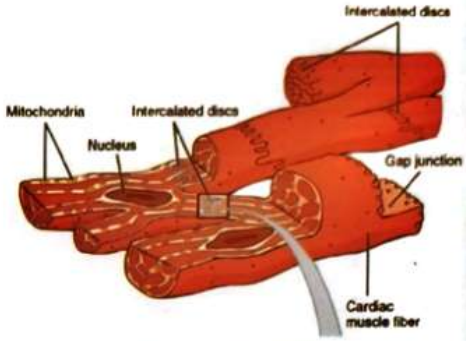
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CARDIAC AND SMOOTH MUSCLES

Cardiac Muscle

00:00:55

Cardiac muscles are found in the heart.



Cardiac muscle

PROPERTIES :

- Large nucleus.
- Plenty of mitochondria → To generate ATP.
- Histology :
  - Presence of intercalated disc containing gap junctions.
  - Gap junctions contains proteins : Connexins.
- Type : Striated muscle.
- Mode of contraction : Involuntary.
  Needs a pacemaker : SA node.
- Regulatory proteins : Troponin & Tropomyosin.
- 2 types of myosin heavy chain :
  α-myosin heavy chain → Atrium.
  β-myosin heavy chain → Ventricles.

Functional Syncytium :

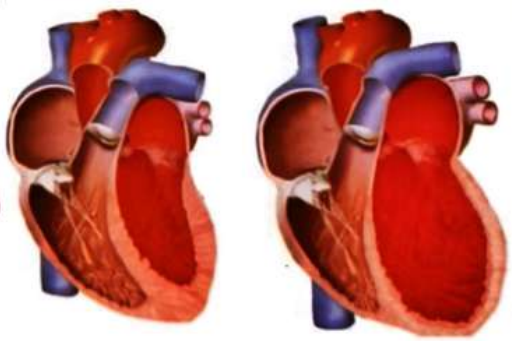
Specific to cardiac muscle.

Presence of gap junction
↓
Rapid transmission of impulses from one cell to another
↓
All cells contract & relax at the same time
↓
Generates large amount of force to pump out blood.

TITIN IN CARDIAC MUSCLE :

- Provides elasticity to cardiac muscle → Ability to recoil.
- Mutation of titin : Dilated cardiomyopathy (DCM).

Muscles stretch without recoil
↓
Dilated cardiac chambers


Normal DCM

UNIQUE FEATURES

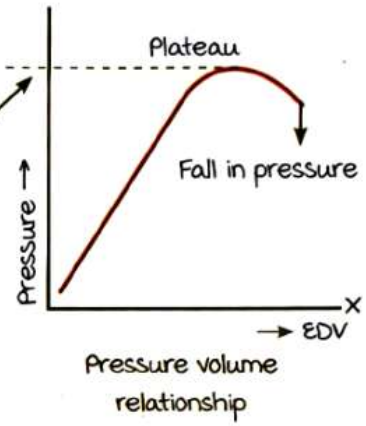
----- Active space -----

Length-tension relationship :

- Blood enters the chambers during diastole : End diastolic volume (EDV).
- EDV → Stretches the chambers → Change in length.
- Tension : D/t force of contraction.

Frank-Starling law :

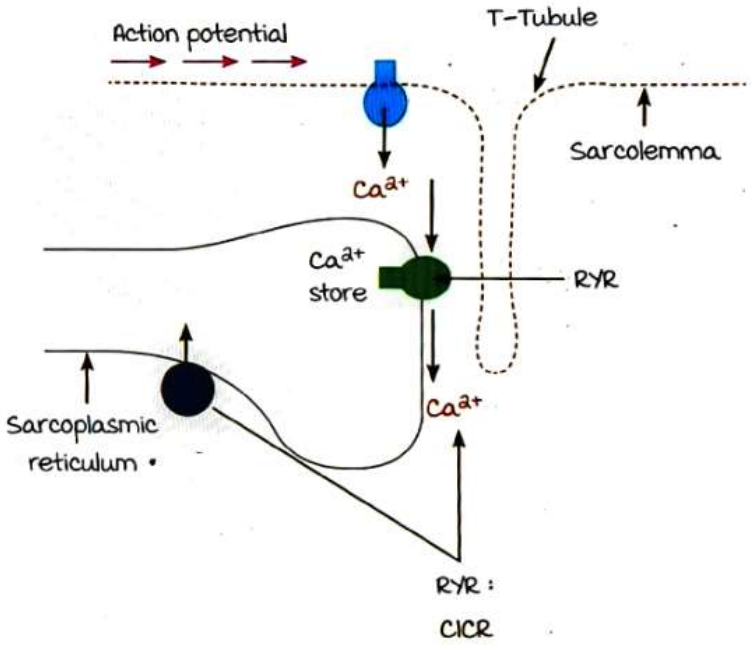
- Force of contraction ∝ Initial length of cardiac muscle fibre within physiological limits.



Pressure volume relationship

Ca^{2+} induced Ca^{2+} release (CICR) :

Action potential
↓
Extracellular Ca^{2+} influx
↓
Release of intracellular Ca^{2+} (iCa^{2+}) from sarcoplasmic reticulum (SR) through RYR
↓
Contraction
↓
iCa^{2+} returns back to SR by SERCA pump
↓
Relaxation



1 T tubule + 1 sarcoplasmic reticulum → Diad arrangement.

Note : In skeletal muscle :

1 T tubule + 2 sarcoplasmic reticulum → Triad arrangement.

Phospholamban and Norepinephrine :

- Phospholamban → (−) SERCA pump.
- NE → (−) Phospholamban (By phosphorylation) → ↑ Ca^{2+} storage.

Norepinephrine : ↑ Ca^{2+} storage by adequate function of SERCA :

- Enhances relaxation → +ve lusitropy.
- Ensures adequate Ca^{2+} for future contraction → +ve inotropy.

Feedback

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----- Active space -----

Absence of tetanisation :

Evolutionary adaptation of cardiac muscle : It cannot be tetanised.

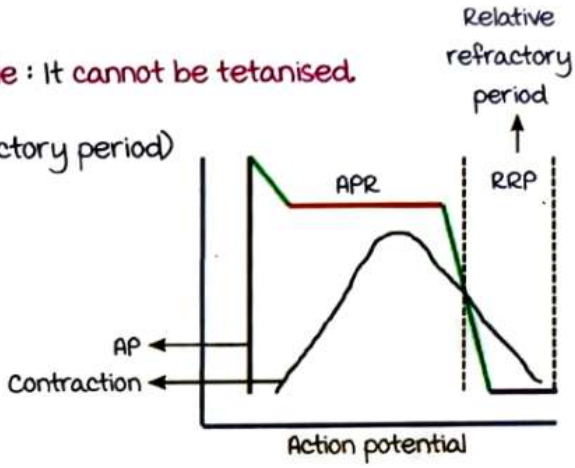
Longer duration of ARP (Absolute refractory period)

↓

No summation possible

↓

No tetany

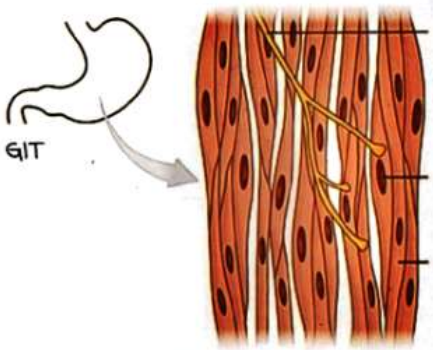


Smooth muscle

00:25:00

PROPERTIES :

- Classical location : GIT.
- Hallmark : Absence of striations.
- Mode of contraction : Involuntary.
- Pacemaker : Cajal cells.
- Dense bodies : Functional equivalent of Z lines.
- Ca^{2+} binding protein : Calmodulin.
- Inelastic d/t absence of titin.



Smooth muscles

TYPES OF SMOOTH MUSCLE :

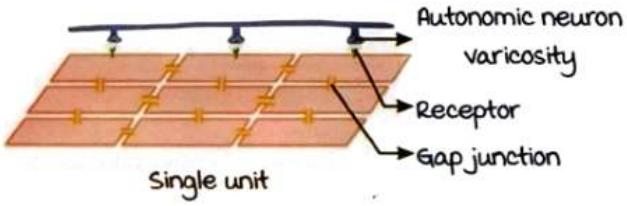
Types

Single unit smooth muscle
- Unitary smooth muscles.
- Gap junctions (+).

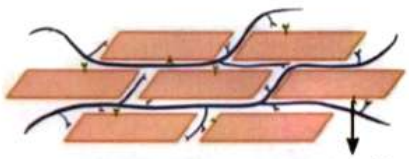
Eg : Uterus, urinary bladder, ureter, GIT.

Multiunit smooth muscle
- Individual subunits.
- No gap junctions.

Eg : Vascular smooth muscle (Blood vessel).



Single unit


Multiunit Varicosities

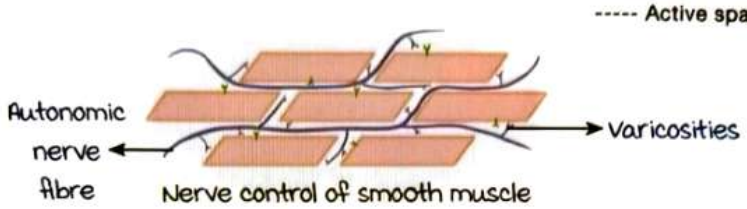
CONTROL OF SMOOTH MUSCLES :

No well-defined neuro-muscular junction regulated by autonomic nervous system.

Feedback

Varicosities :

- Enlargement of nerve endings.
- Contain neurotransmitters :
  - Acetylcholine → Parasympathetic.
  - Norepinephrine → Sympathetic.



Synapse en passant :

One nerve fibre → Innervates multiple smooth muscle cells.

CONTRACTION & RELAXATION :

Parasympathetic nervous system :

- ↑Ca^{2+} in the cell → Contraction.

Source of Ca^{2+} → Extracellular.
→ Intracellular : Released through → • RYR.
• IP_3 receptor.

Mechanism of contraction : Ca^{2+} + calmodulin (Ca^{2+} calmodulin complex)

↓

Activates myosin light chain Kinase (MLCK)

↓

Activates myosin

↓

myosin actin overlap

Mechanism of relaxation : By activation of myosin light chain phosphatase (MLCP).

Note :-

- Rho Kinase → Inhibits MLCP → Inhibits relaxation.
- Fasudil → Rho Kinase → (+) MLCP → Relaxation.
  (−)

Sympathetic nervous system :

Inhibition of MLCK → Inhibition of contraction.

UNIQUE FEATURES :

Plasticity :

Smooth muscles lack titin → Plasticity (Stretch without recoil).

Eg : Uterus keeps stretching without recoil as fetus grows.
Urinary bladder stretches to store urine.

Latch bridge phenomenon :

Energy efficiency of smooth muscle.

Sustained contraction with low ATP.

Eg: Sustained contraction of GIT for digestion.

Feedback